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DEEP LEARNING-BASED NETWORK TRAFFIC CLASSIFICATION

Dynamic QoS Management in SDNs





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Agenda

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Background

- SDN enables centralized and programmable network management.
- Modern traffic includes VoIP, Video Streaming, File Transfer, and Web Browsing.
- Different traffic types require different QoS policies.
- Traditional methods (Port-based, DPI, Rule-based) are becoming ineffective.
- Deep Learning can classify traffic automatically for dynamic QoS.

Related Work & Research Gap

Four Categories of Related Studies:

- DL-based Traffic Classification in SDN RF, GRU, GRU, LSTM, and CNN-LSTM models evaluated on VPN-nonVPN and InSDN datasets. [1][2][8]
- DL/ML-driven Dynamic QoS Flow Persona, OpenFlow meters, and prioritized traffic engineering for QoS optimization. [3][4][6]
- Mininet/Ryu Deployment & Controller Overhead Evaluation of controller CPU usage, memory consumption, and deployment overhead. [5][13]
- Survey Papers Comprehensive reviews of challenges and future directions in QoS-aware ML/DL for SDN. [7]

Research Gap

1. Most studies are evaluated offline on static datasets.
2. QoS mechanisms rely on simple rule-based methods rather than DL-based classification.
3. Few studies jointly evaluate classification accuracy, QoS improvement, and controller overhead.
4. CNN-GRU remains underexplored for real-time multi-class traffic classification in SDN.

RESEARCH QUESTIONS

RQ1
How accurately can the proposed CNN-GRU classify multi-class network traffic (VoIP, Video, File, Web) compared with existing ML/DL models?

RQ2
To what extent does integrating the classification module into the SDN controller improve QoS performance?

RQ3
What is the controller overhead of deploying the DL model in a real-time SDN environment?

- Obj 1.1: Develop a CNN-GRU classifier using SDN flow statistics.
- Obj 1.2: Compare with RF, CNN, LSTM, and GRU using Accuracy, Precision, Recall, F1-score, and AUC-ROC.

- Obj 2.1: Design a dynamic QoS mechanism based on traffic classification.
- Obj 2.2: Evaluate throughput, latency, and packet loss in Mininet + Ryu.

- Obj 3.1: Measure classification latency, CPU, and RAM usage.
- Obj 3.2: Compare controller overhead with a rule-based baseline.

METHODOLOGY

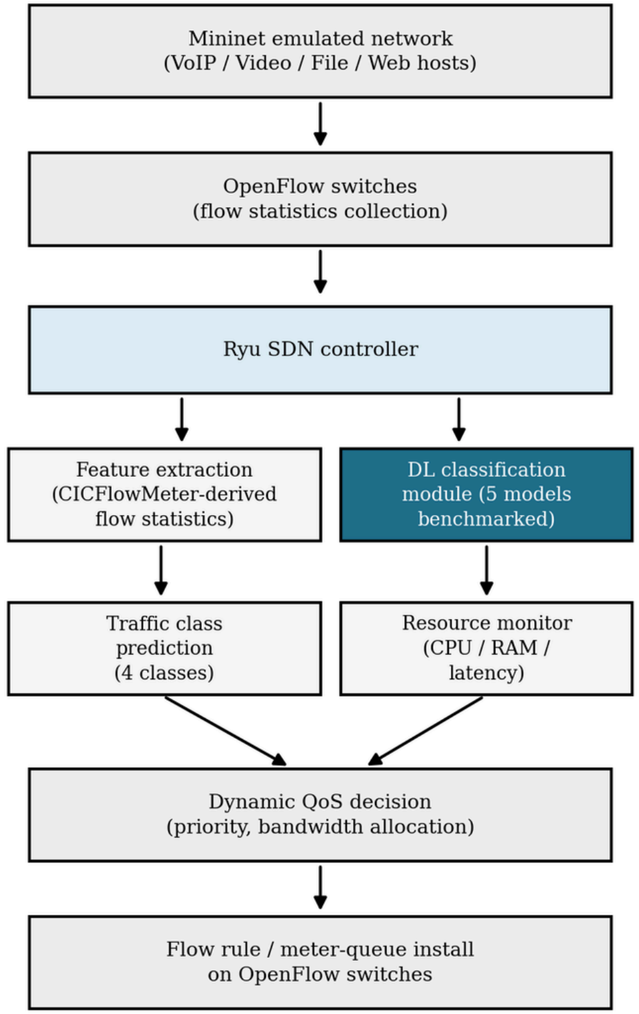
Data Plane: Mininet network generating mixed traffic (4 classes).

Control Plane: Ryu SDN controller collecting flow statistics.

Classification Module: AI evaluates flows and outputs predictions.

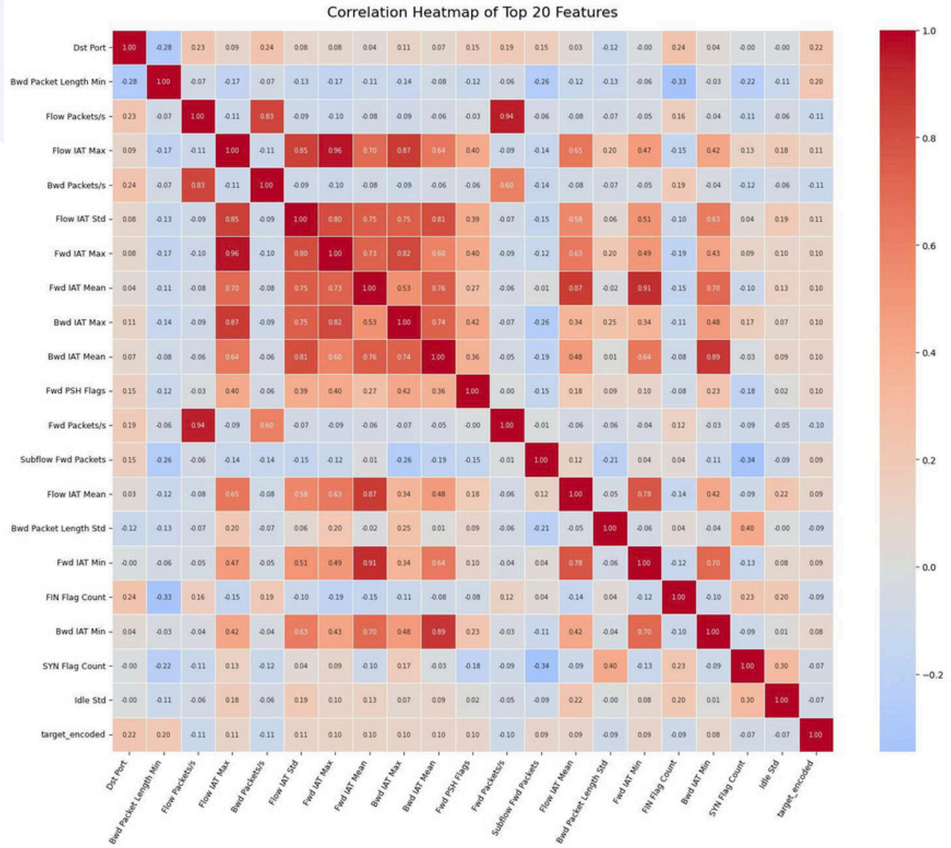
QoS Enforcement: Controller automatically assigns priority and bandwidth based on AI predictions.

System Architecture of the CNN-GRU-based Ryu Controller



EXPERIMENTS

Correlation Heatmap



ISCX VPN-nonVPN Dataset

- 4 traffic classes: VoIP, Video, File Transfer, Web.
- Features: CICFlowMeter flow statistics (duration, packets, bytes, IAT, packet size, flow rate).
- Preprocessing: Standardization, one-hot encoding, class weighting.
- Data split: 70% Train • 15% Validation • 15% Test.

EXPERIMENTS

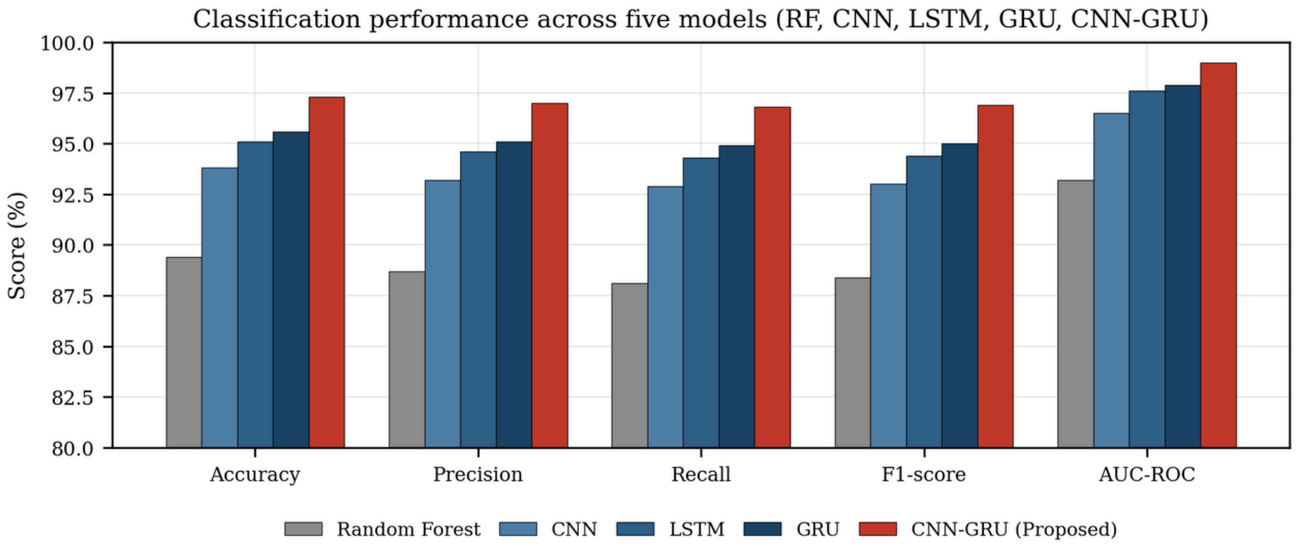
Experimental Environment

Tools & Environment

- Mininet + Open vSwitch
  OpenFlow 1.3
- Ryu SDN Controller
  Python-based controller with host classification and QoS modules
- TensorFlow / Keras
  Model training and inference
- iPerf3 / HTTP Traffic Generator
  Generates VoIP, video, file transfer, and web traffic
- ping / hping3
  Latency measurement
- psutil
  Monitors CPU and RAM usage of the controller process

Results

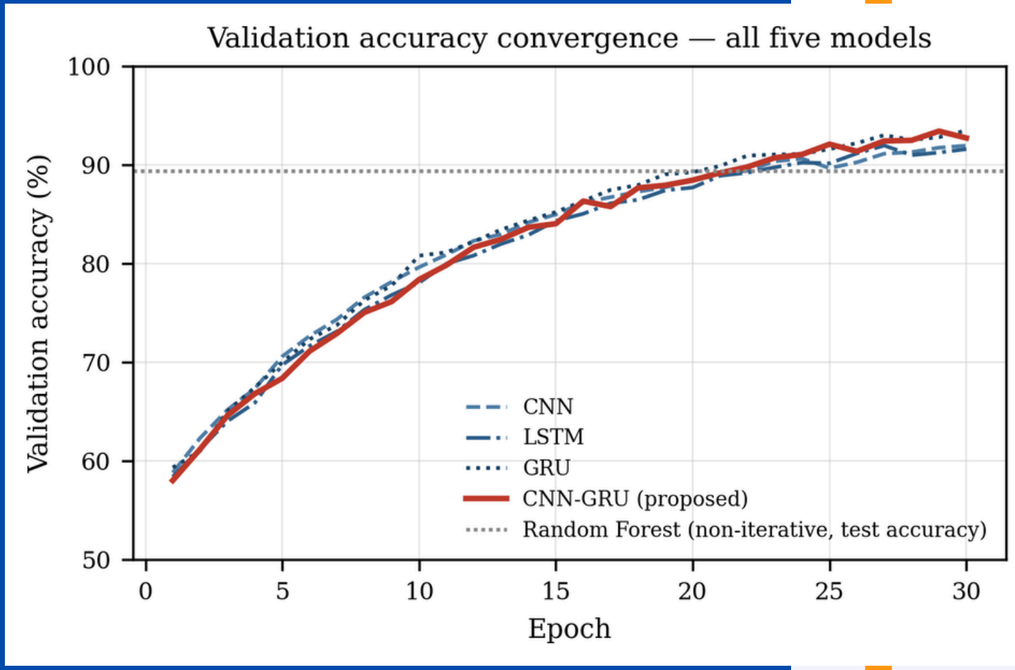
Classification Performance(RQ1)



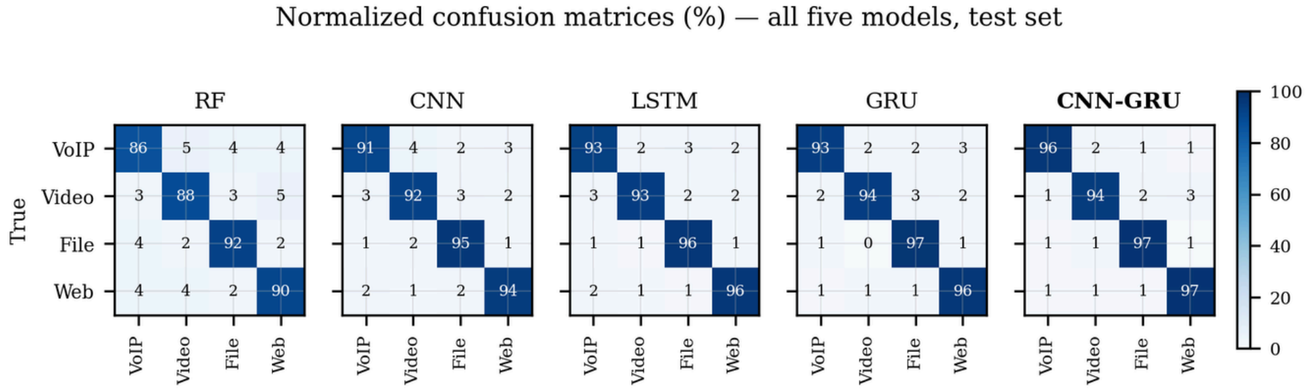
- The proposed CNN-GRU model achieves 97.3% accuracy.
- Spatial features captured by CNN layers; temporal flow dependencies captured by GRU layers.
- Outperforms all baselines: Random Forest (89.4%), CNN, LSTM, and standalone GRU (95.6%).

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Model Robustness & Convergence



- High Precision: Achieves >94% per-class recall across all four traffic categories.
- Rapid Convergence: CNN-GRU reaches stable, high accuracy after approximately 20 epochs of training.
- Successfully distinguishes difficult bursty traffic patterns (e.g., Video vs. Web).



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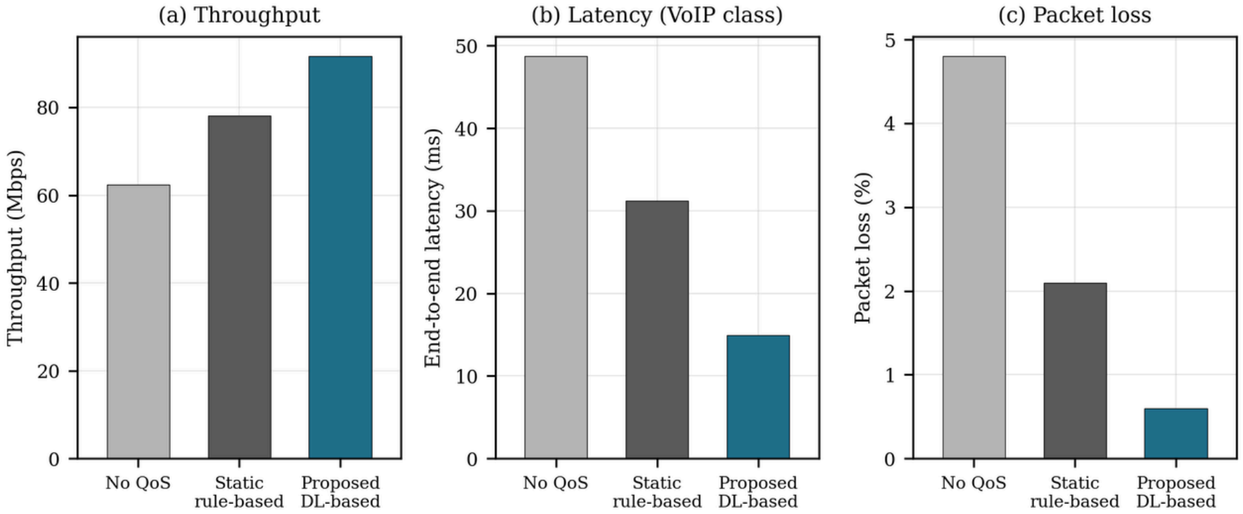
Results

Dynamic QoS Impact (RQ2)

Throughput: Improved by ~47% (from 62.4 to 91.6 Mbps) compared to best-effort forwarding.

Latency: End-to-end VoIP latency reduced by ~69% (48.7ms down to 14.9ms).

Packet Loss: Dropped significantly from 4.8% to just 0.6%.



QoS configurations: No QoS = best-effort forwarding; Static = static rule-based QoS; Proposed = DL-based dynamic QoS (CNN-GRU-driven).

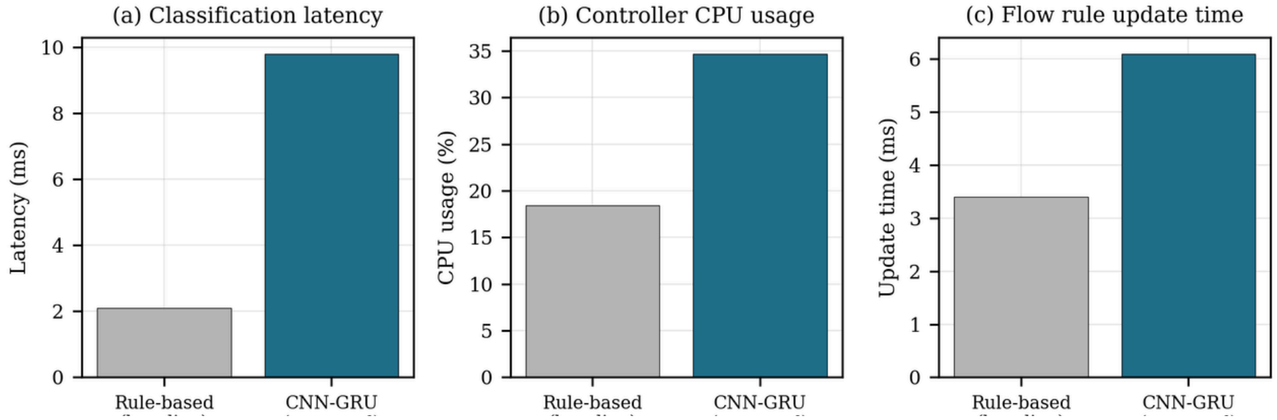
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Results

Controller Overhead (RQ3)

The Trade-off: DL classification increases CPU usage (34.7%) and latency (9.8ms) compared to static rules.

Practical Feasibility: Despite the overhead, metrics remain within the acceptable sub-10ms range for openFlow deployments.



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Discussion & Limitations

Trade-off: Accuracy vs. Complexity

- Best for performance-critical QoS, where high classification accuracy is required.
- Accuracy–complexity trade-off: CNN-GRU achieves the highest accuracy but with higher latency and training cost.
- GRU is a lightweight alternative for resource-constrained SDN controllers.
- Future work: Optimize inference speed and evaluate on larger real-world networks.

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Discussion & Limitations

- Note on Results: The reported figures are illustrative values used to validate the evaluation framework prior to full empirical data collection.
- External Validity: Emulated Mininet topology operates at a small scale and uses synthetic traffic, which may not fully represent production networks.
- Construct Validity: The 4-class taxonomy is a simplification; encrypted/tunneled traffic edge cases require further profiling.

Conclusion & Future Work

- Deep Learning embedded in SDN controllers is highly effective for dynamic, real-time QoS management.
- The CNN-GRU hybrid model offers the best balance of accuracy and temporal awareness.
- Pooling layers reduce dimensionality while preserving dominant spatial features.
- Future Work: Scaling to physical hardware, model quantization, and expanding to encrypted-traffic edge cases.

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Thank You For Your Listening